QP CODE: 25901815

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCZGY101 - BIOLOGICAL BASIS OF BEHAVIOR II

(2024 Admission onwards)

Duration**: 1 Hour 30 Minutes** Maximum Marks: **50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** questions
Each question carries **1** mark

1. What is the first step in cell communication?
   - c) A signal binds to a receptor
2. The process where homologous chromosomes pair up is called ____.
   - b) Synapsis
3. G1 and G2 phase of cell cycle are:
   - a) Growth phase
4. In a monohybrid cross involving a heterozygous pair (Tt × Tt) where T is dominant for tall and t is recessive for short, what is the expected phenotypic ratio in the offspring?
   - c) 3 tall : 1 short
5. Which of the following is a sex-linked disorder?
   - a) Hemophilia
6. The next step after transcription is:
   - c) Translation
7. Who discovered the double-helix structure of DNA?
   - a) Watson and Crick
8. Which of the following is a feature of Cri-du-chat syndrome?
   - a) Cat-like cry

9. What is the primary role of genetic counselling?
    - b) To provide information and guidance to patients and families
10. Karyotyping is used for:
    - b) Chromosome analysis **(10 x 1 = 10 Marks)**

Part B

Short Answer Questions

Answer any **10** questions
Each question carries **2** marks

11. What is nucleolus?

- The Nucleolus is a dense structure present inside the nucleus of eukaryotic cells.
- It is the site of ribosomal RNA (rRNA) synthesis.
- It helps in the formation of ribosomes. (Any 2 points)

12. Comment on the role of hormones as signalling molecules.

- Hormones are chemical messengers secreted by endocrine glands.
- They travel through blood to target cells.
- They regulate growth, metabolism, reproduction, and homeostasis.(definition: 0.5 mark, Any 2 roles: 1.5 mark)

13. What is the key characteristic of paracrine signaling?

- Paracrine Signaling acts on nearby cells.
- Chemical signals diffuse over short distances.
- It is important in tissue repair and immune responses.(definition: 1 mark, Mechanism: 1 mark)

14. What is the phenotypic ratio and genotypic ratio of a dihybrid cross?

- In a dihybrid cross (AaBb × AaBb):
- Phenotypic ratio = **9 : 3 : 3 : 1**
- Genotypic ratio = **1:2:1:2:4:2:1:2:1** (1 mark each for the ratios)

15. What is the relevance of DNA replication?

- DNA Replication ensures genetic continuity.
- It produces identical copies of DNA before cell division.
- It is essential for growth and repair.
- It helps in transmission of hereditary information.
- It occurs during the S phase of the cell cycle. (Any 2 functions)

16. How does a test cross help in identifying the genotype of an individual?

- A test cross involves crossing with a homozygous recessive individual.
- It helps identify whether the dominant individual is homozygous or heterozygous.
- All dominant offspring indicate homozygous dominant genotype.
- Appearance of recessive offspring indicates heterozygous genotype.
- It is widely used in genetics and breeding experiments.(Definition or explain with a cross)

17. Explain the key points of chromosome theory of inheritance.

- Chromosome Theory of Inheritance states that genes are located on chromosomes.

- Chromosomes occur in pairs in diploid organisms.
- Chromosomes segregate during meiosis.
- It explains Mendelian inheritance patterns.
- Sutton and Boveri proposed this theory.

18. What is Turner's syndrome?

- Turner Syndrome is caused by the absence of one X chromosome (45,X).
- It occurs only in females.
- Affected individuals are usually short in stature.
- They may have infertility due to underdeveloped ovaries.
- Delayed puberty is a common feature. (Karyotype :0.5mark, symptoms: 1.5 mark)

19. What is the significance of identifying consanguineous marriages in a pedigree analysis?

- Consanguineous Marriage refers to a marriage between individuals who are biologically related, such as cousins.
- Pedigree Analysis helps trace inherited disorders.
- Consanguineous marriages increase recessive disorders.
- It helps in identifying carrier individuals.
- Useful in genetic counselling and risk prediction.
- It assists in understanding inheritance patterns in families. (Any 2 points: 2 mark)

20. Name two types of numerical chromosomal aberrations.

- Two types are: Aneuploidy and Polyploidy.
- Aneuploidy involves loss or gain of chromosomes.
- Polyploidy involves extra sets of chromosomes.

21. What is spontaneous mutation?

- Spontaneous mutation refers to a naturally occurring change in an organism's DNA sequence that arises without external influence from mutagens such as chemicals or radiation. These mutations are a fundamental source of genetic variation and play a central role in evolution, heredity, and disease.
- It may arise due to errors in DNA replication.
- It contributes to genetic variation. (Any 2 points)

22. What is eugenics? Give an example.

- Eugenics is the concept of improving the genetic quality of a human population through selective breeding or controlled reproduction.
- It aims to reduce hereditary disorders.
- Example: discouraging marriages between carriers of severe genetic diseases.
- It was historically applied in population improvement programs.
- Modern genetics emphasizes ethical concerns regarding eugenics. (definition: 1 mark, Example : 1 mark) **(10 x 2 = 20 Marks)**

Part C

Short Essay Questions

Answer any **5** questions
Each question carries **4** marks

23. Describe the main characteristics of eukaryotic cells?

- Eukaryotic Cell are complex cells found in plants, animals, fungi, and protists.
- Presence of a true nucleus enclosed by nuclear membrane.
- Membrane-bound organelles like mitochondria, ER, and Golgi body are present.
- DNA is linear and arranged as chromosomes.
- Cells are larger and more complex than prokaryotic cells.
- 80S ribosomes are present.
- Cell division occurs by mitosis and meiosis.
- Cytoskeleton helps maintain shape and movement.
- Plasma membrane regulates movement of substances. (A short note on the characters)

24. Explain anaphase in mitosis with the help of a diagram.

- Anaphase is the stage of mitosis in which sister chromatids separate.
- Centromeres split into two.
- Sister chromatids move to opposite poles.
- Spindle fibres help chromosome movement.
- Each chromatid becomes a daughter chromosome.
- Equal distribution of chromosomes occurs. (Explanation of the events: 2 marks, diagram: 2 marks)

25. State and explain Mendel's Law of Segregation with an example.

- Mendel's Law of Segregation is a foundational principle of genetics proposed by Gregor Mendel. It states that during gamete formation, the two alleles for a given trait separate so that each gamete carries only one allele. This explains how offspring can inherit different genetic combinations from their parents.

- **Example**

  Tall pea plant cross: Tt×Tt

  Genotypic ratio: 1TT:2Tt:1tt

  Phenotypic ratio: 3 Tall:1 Dwarf (Mendels law: 1.5 mark, Example: 2.5 mark)

26. Explain the process of transcription.

- Transcription is the process by which RNA is synthesized from a DNA template. It is the first step of gene expression.
- Template: DNA
- Main enzyme: RNA polymerase
- Product: RNA (mRNA, tRNA, rRNA)
- Occurs in nucleus of eukaryotes and cytoplasm of prokaryotes

- RNA synthesis occurs in 5′ → 3′ direction
- **Steps of transcription**

1. **Initiation** – RNA polymerase binds to promoter region.
2. **Elongation** – Complementary RNA nucleotides are added.
3. **Termination** – RNA strand is released at stop signal.

- Transfers genetic information from DNA to RNA.
- Essential for protein synthesis and gene expression. (A detailed account: 4 marks)

27. Differentiate between phenotype and genotype.

Phenotype	Genotype
Observable characteristics of an organism	Genetic constitution of an organism
Influenced by genes and environment	Determined mainly by genes
Influenced by genes and environment	Determined mainly by genes
Can be physically observed	Cannot be directly observed
Example: Tall plant	Example: TT or Tt

(Any 4 differences)

28. How can point mutations contribute to genetic diseases? Give an example.

- Point Mutation is a change in one nucleotide of DNA.
- Alters codon sequence.
- Produces abnormal proteins.
- Can lead to genetic diseases.
- Eg:Sickle Cell anemia
- Caused by mutation in beta-globin gene.
- RBCs become sickle-shaped.
- Leads to anemia and poor oxygen transport. (Description: 3 mark, example; 1mark)

29. Define euthenics and its significance in improving the current human population.

- Euthenics is the study and practice of improving human well-being and performance through the enhancement of environmental conditions.
- Improves health and nutrition.
- Reduces diseases and mortality.
- Enhances education and living standards.
- Increases productivity and social welfare.
- Helps national development. (definition: 1 mark, significance: 3 mark)

(5 x 4 = 20 Marks)

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